

DLFA Class Test 2

Question 2

$$1. PE(i) = \begin{bmatrix} \sin(2\pi f_0 i) \\ \cos(2\pi f_0 i) \\ \sin(2\pi f_1 i) \\ \cos(2\pi f_1 i) \\ \vdots \\ \vdots \\ \sin(2\pi f_{G3} i) \\ \cos(2\pi f_{G3} i) \end{bmatrix}$$

$$PE(i + \delta) = \begin{bmatrix} \sin(2\pi f_0 (i + \delta)) \\ \cos(2\pi f_0 (i + \delta)) \\ \sin(2\pi f_1 (i + \delta)) \\ \cos(2\pi f_1 (i + \delta)) \\ \vdots \\ \vdots \\ \sin(2\pi f_k (i + \delta)) \\ \cos(2\pi f_k (i + \delta)) \end{bmatrix}$$

$$PE(i+L) = \begin{bmatrix} \sin(2\pi f_0 i) \cos(2\pi f_0 \delta) + \cos(2\pi f_0 i) \sin(2\pi f_0 \delta) \\ \cos(2\pi f_0 i) \cos(2\pi f_0 \delta) - \sin(2\pi f_0 i) \sin(2\pi f_0 \delta) \\ \sin(2\pi f_1 i) \cos(2\pi f_1 \delta) + \cos(2\pi f_1 i) \sin(2\pi f_1 \delta) \\ \cos(2\pi f_1 i) \cos(2\pi f_1 \delta) - \sin(2\pi f_1 i) \sin(2\pi f_1 \delta) \\ \vdots \\ \sin(2\pi f_N i) \cos(2\pi f_N \delta) + \cos(2\pi f_N i) \sin(2\pi f_N \delta) \\ \cos(2\pi f_N i) \cos(2\pi f_N \delta) - \sin(2\pi f_N i) \sin(2\pi f_N \delta) \end{bmatrix}$$

$$= \begin{bmatrix} \cos(2\pi f_0 \delta) & \sin(2\pi f_0 \delta) & 0 & 0 & \dots & 0 \\ 0 & 0 & \cos(2\pi f_0 \delta) - \sin(2\pi f_0 \delta) & & & \\ & & & & & \\ & & & & & \\ 0 & 0 & & & \cos(2\pi f_N \delta) - \sin(2\pi f_N \delta) & \\ & & & & & \end{bmatrix}$$

last 2 columns

$$X \begin{bmatrix} \sin 2\pi f_0 i \\ \cos 2\pi f_0 i \\ \vdots \\ \sin 2\pi f_n i \\ \cos 2\pi f_n i \end{bmatrix}_{128 \times 1}$$

$$= T(S)_{128 \times 128} PE(i)_{128 \times 1}$$

2. The positional encoding at $i+S$ is a linear transformation of the positional encoding at i , where the linear transformation depends only on the offset S between the two positions. As the change in the positional encode only depends on the offset between the two positions, this positional encoding scheme is capable of capturing the relative positions of the tokens of the sequence to which it is applied.

Question 3

$$X = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \\ 2 & 1 & 0 \end{bmatrix}$$

(input)

$$W = \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix}$$

(filter weights)

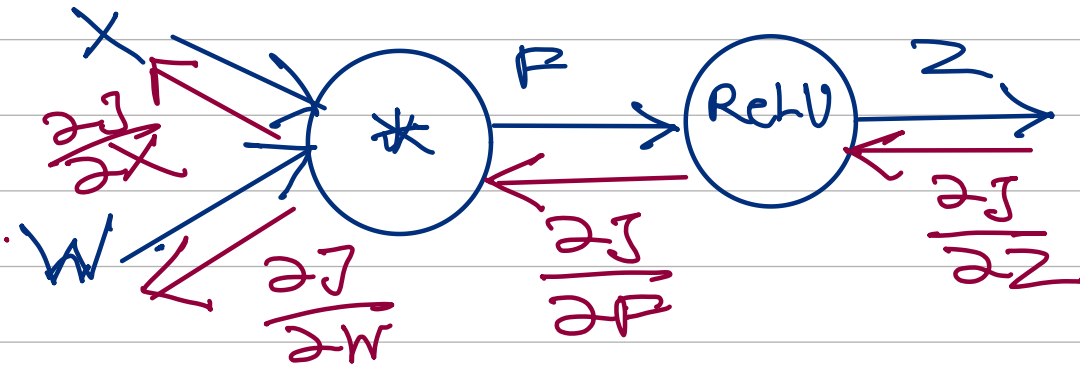
$$\underline{1.} \quad X * W = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \\ 2 & 1 & 0 \end{bmatrix} * \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix}$$
$$= \begin{bmatrix} 1 & 7 \\ 1 & -2 \end{bmatrix}$$

$$Z = \text{ReLU}(X * W)$$

$$= \begin{bmatrix} 1 & 7 \\ 1 & 0 \end{bmatrix}$$

$$2) \frac{\partial J}{\partial Z} = \begin{bmatrix} 1 & -2 \\ 2 & 3 \end{bmatrix}$$

let $P = X * W$, $Z = \text{ReLU}(P)$



$$Z = \text{ReLU}(P)$$

$$\frac{\partial J}{\partial P} = \frac{\partial J}{\partial Z} \cdot \frac{\partial Z}{\partial P}$$

$$= \frac{\partial J}{\partial Z} \odot \text{ReLU}'(P)$$

$$= \begin{bmatrix} 1 & -2 \\ 2 & 3 \end{bmatrix} \odot \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -2 \\ 2 & 0 \end{bmatrix}$$

$$\frac{\partial J}{\partial W} = X * \frac{\partial J}{\partial F}$$

$$= \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \\ 2 & 1 & 0 \end{bmatrix} * \begin{bmatrix} 1 & -2 \\ 2 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} -3 & 2 \\ 2 & -3 \end{bmatrix}$$

$$3) \frac{\partial J}{\partial F} = \begin{bmatrix} 0 & 2 \\ -2 & 1 \end{bmatrix} \text{ (180° rotated)}$$

$$\frac{\partial J}{\partial X} = W * \frac{\partial J}{\partial F}$$

↑
full convolution

$$= \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix} * \begin{bmatrix} 0 & 2 \\ -2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -3 & 2 \\ 2 & 0 & -4 \\ 0 & 4 & 0 \end{bmatrix}$$

Question 4

$$1. K = 2, W_i = 2, S = 1, P = 0$$

$$W_o = K + (W_i - 1) \times S - 2P$$

$$= 2 + (2 - 1) \times 1$$

$$= 2 + 1$$

$$= 3$$

So, the dimension of the transposed convolution result will be 3×3 .

2. Size of convolution matrix M will be 4×9

$$\frac{3}{1} M = \begin{bmatrix} 1 & 0 & -0 & -1 & 2 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & -1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & -1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Toeplitz Blocks :

$$B_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, B_2 = \begin{bmatrix} -1 & 2 & 0 \\ 0 & -1 & 2 \end{bmatrix}$$

$$0 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M = \begin{bmatrix} B_1 & B_2 & 0 \\ 0 & B_1 & B_2 \end{bmatrix}$$

$$\underline{4.} \quad \bar{Z} = M^T X$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 2 & -1 & 0 & 1 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 2 & -1 \\ 0 & 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 2 \\ 0 \\ 2 \\ 4 \\ 4 \\ -3 \\ 2 \\ 8 \end{bmatrix}$$

$$\therefore Z = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 4 & 4 \\ -3 & 2 & 8 \end{bmatrix}$$

Question 1

1. B, C

2. B, C

3. C.

Each GRU - layer = 2 gates

Total gates = $8 \times 2 \times 2 = 32$